SSL Certificate Tool

Generate on Linux Server (Script Only)

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Use this guide when Windows/Mac GUI packages are not available. Generate Ameyo-ready SSL files (PEM, P12, JKS, Asterisk, ca-bundle) directly on a Linux server.

1. Why no Mac ZIP on Windows?

Mac applications must be built on a Mac computer. Running build-windows.bat only creates SSLCertTool-Windows.zip. To create Mac ZIP, on a Mac run: ./build-mac.sh

Package matrix

- Windows ZIP: build-windows.bat on Windows
- Linux Source ZIP: pack-linux-source.bat on Windows (uses python3 on server)
- Linux Binary ZIP: ./build-linux.sh on Linux
- Mac ZIP: ./build-mac.sh on Mac only

2. Copy Linux source package to server

```
scp SSLCertTool-Linux-Source.zip dacx@server:/dacx/
ssh dacx@server
cd /dacx
unzip SSLCertTool-Linux-Source.zip -d SSLCertTool
cd SSLCertTool
chmod +x generate-on-linux.sh
```

3. One-time setup

```
# RHEL/CentOS
sudo yum install -y python3 python3-pip
# Ubuntu/Debian
sudo apt install -y python3 python3-pip

python3 -m pip install --user -r requirements.txt
# Offline: cd offline && ./install-offline.sh
```

4. Generate certificates

4.1 Interactive

```
./generate-on-linux.sh
```

4.2 One command

```
./generate-on-linux.sh \
  --input /dacx/SSL2026/customer.eml \
  --key /dacx/SSL2026/domain.key \
  --password 'YOUR_KEYSTORE_PASSWORD' \
  --alias ameyo-server \
  -o /dacx/SSL2026/out
```

4.3 Using ssl-auto.py directly

```
python3 ssl-auto.py generate \
  --input /dacx/SSL2026/customer.eml \
  --key /dacx/SSL2026/domain.key \
  --password 'YOUR_KEYSTORE_PASSWORD' \
  --alias ameyo-server \
  -o /dacx/SSL2026/out
```

Password must match connector.https.keystorePass on Tomcat. Alias is usually ameyo-server.

5. Output files

- main_<domain>.crt - CRM/Apache leaf (often rename to crm.crt)
- <domain>.key - private key
- chain.crt / crm-ca.crt - Apache chain
- <domain>.ca-bundle - legacy ca-bundle (intermediate + root)
- <domain>.pem - full chain (certs only)
- asterisk.pem - KEY first + certs (required for Ameyo WSS)
- asterisk_<domain>.crt - Asterisk cert chain
- <domain>.jks - Tomcat JKS (needs Java keytool)
- keystore.p12 - PKCS12

6. Verify

```
openssl x509 -noout -modulus -in out/*/main_*.crt | openssl md5
openssl rsa  -noout -modulus -in out/*/*.key | openssl md5
grep -E 'BEGIN PRIVATE KEY|BEGIN CERTIFICATE' out/*/asterisk.pem | head
keytool -list -keystore out/*/*.jks -storepass 'PASS' -alias ameyo-server
```

7. Deploy (examples)

```
# Backup
cp /dacx/ssl2025/app.jks /dacx/ssl2025/app.jks.bak.$(date +%Y%m%d)

# Tomcat JKS (path = connector.https.keystoreFile)
cp out/<domain>/<domain>.jks /dacx/ssl2025/app.jks

# CRM
cp out/<domain>/main_<domain>.crt /dacx/ssl2025/crm.crt
cp out/<domain>/<domain>.key /dacx/ssl2025/<domain>.key
cp out/<domain>/chain.crt /dacx/ssl2025/chain.crt

# Asterisk
KEYS=/dacx/var/ameyo/dacxdata/asterisks/13/etc/asterisk/keys
cp out/<domain>/asterisk.pem $KEYS/asterisk.pem
cp out/<domain>/<domain>.key $KEYS/asterisk.key
chmod 600 $KEYS/asterisk.key $KEYS/asterisk.pem
# then restart AppServer / httpd / asterisk
```

8. JKS import on Linux (if needed)

```
keytool -importkeystore \
  -srckeystore /dacx/SSL2026/keystore.p12 \
  -srcstoretype PKCS12 \
  -srcstorepass 'PASS' \
  -destkeystore /dacx/ssl2025/app.jks \
  -deststoretype JKS \
  -deststorepass 'PASS' \
  -srcalias ameyo-server \
  -destalias ameyo-server \
  -noprompt
```

Do NOT import into .../cacerts for Tomcat HTTPS. Copy/import into the path from connector.https.keystoreFile. Default cacerts password is changeit (unrelated).

9. Checklist

- [ ] Unzip Linux source package on server
- [ ] Install python3 + pip requirements
- [ ] Have customer cert + matching private key
- [ ] Generate with correct keystore password/alias
- [ ] Confirm Key/Cert match and asterisk.pem KEY-first
- [ ] Backup old server certs
- [ ] Copy to Tomcat/CRM/Asterisk paths
- [ ] Restart and verify HTTPS + WSS 8089